Data Science Project

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Introduction

The purpose of this project is to gauge your technical skills and problem solving ability by working through something similar to a real NBA data science project. You will work your way through this R Markdown document, answering questions as you go along. Please begin by adding your name to the "author" key in the YAML header. When you're finished with the document, come back and type your answers into the answer key at the top. Please leave all your work below and have your answers where indicated below as well. Please note that we will be reviewing your code so make it clear, concise and avoid long printouts. Feel free to add in as many new code chunks as you'd like.

Remember that we will be grading the quality of your code and visuals alongside the correctness of your answers. Please try to use the tidyverse as much as possible (instead of base R and explicit loops). Please do not bring in any outside data.

Note:

Throughout this document, any `season` column represents the year each season started. For example, the 2015-16 season will be in the dataset as 2015. For most of the rest of the project, we will refer to a season by just this number (e.g. 2015) instead of the full text (e.g. 2015-16).

Answers

Part 1

Question 1:

-Offensive: 56.3% eFG
-Defensive: 47.9% eFG

Question 2: 81.6%

Question 3: 46.2%

Question 4: This is a written question. Please leave your response in the document under Question 5.

Question 5: XX.X% of games

Question 6:

- Round 1: XX.X%
- Round 2: XX.X%
- Conference Finals: XX.X%

- Finals: XX.X%

Question 7:

- Percent of +5.0 net rating teams making the 2nd round next year: XX.X%
- Percent of top 5 minutes played players who played in those 2nd round series: XX.X%

Part 2

Please show your work in the document, you don't need anything here.

Part 3

Please write your response in the document, you don't need anything here.

Setup and Data

```
library(tidyverse)
# Make sure we are in our project directory
setwd("/Users/kevinrouse78/Desktop/Thunder_Project")
# Read in data
player_data <- read_csv("Data/player_game_data.csv")
team_data <- read_csv("Data/team_game_data.csv")
```

Part 1 – Data Cleaning

In this section, you're going to work to answer questions using data from both team and player stats. All provided stats are on the game level.

Before working with the data, I like to check for missingness levels among the variables

```
# Check for missing values in the team_data dataset
team_missing_values <- team_data %>%
  summarise_all(~sum(is.na(.)))

# Display columns with missing values and the number of missing values
team_missing_values %>%
  gather(key = "Column", value = "MissingValues") %>%
  filter(MissingValues > 0)
```

```
## # A tibble: 0 x 2
## # i 2 variables: Column <chr>, MissingValues <int>
```

```r
# Do the same for the player_data
# Check for missing values in the player_data dataset
player_missing_values <- player_data %>%
  summarise_all(~sum(is.na(.)))

# Display columns with missing values and the number of missing values
player_missing_values %>%
  gather(key = "Column", value = "MissingValues") %>%
  filter(MissingValues > 0)
```

```
## # A tibble: 1 x 2
##   Column      MissingValues
##   <chr>               <int>
## 1 player_name             4
```

We have 4 missing values in the player name column in the player data

```r
# filter out to get missing players
missing_players <- player_data %>%
  filter(is.na(player_name))
print(missing_players)
```

```
## # A tibble: 4 x 59
##   nbagameid gamedate   season gametype nbapersonid player_name  nbateamid team
##       <dbl> <date>      <dbl>    <dbl>       <dbl> <chr>            <dbl> <chr>
## 1  40900131 2010-04-17   2009        4     3216458 <NA>        1610612738 BOS
## 2  21400621 2015-01-19   2014        2      203296 <NA>        1610612758 SAC
## 3  21500512 2016-01-04   2015        2        6274 <NA>        1610612755 PHI
## 4  21701147 2018-04-01   2017        2  1962937848 <NA>        1610612737 ATL
## # i 51 more variables: team_name <chr>, opposingnbateamid <dbl>,
## #   opp_team <chr>, opp_team_name <chr>, starter <dbl>, missed <dbl>,
## #   seconds <dbl>, points <dbl>, fg2made <dbl>, fg2missed <dbl>,
## #   fg2attempted <dbl>, fg3made <dbl>, fg3missed <dbl>, fg3attempted <dbl>,
## #   fgmade <dbl>, fgmissed <dbl>, fgattempted <dbl>, ftmade <dbl>,
## #   ftmissed <dbl>, ftattempted <dbl>, reboffensive <dbl>, rebdefensive <dbl>,
## #   offensivereboundchances <dbl>, defensivereboundchances <dbl>, ...
```

We can see that these players have their nbapersonid's so we will retreive their names and add them

```r
# make a player reference with each players name and ID
player_reference <- player_data %>%
  filter(!is.na(player_name)) %>%
  select(nbapersonid, player_name) %>%
  distinct()
# Join the missing player names with the reference dataset to get the missing player names
updated_missing_players <- missing_players %>%
  left_join(player_reference, by = "nbapersonid") %>%
  select(-player_name.x) %>%
  rename(player_name = player_name.y)

# Display the updated names
print(updated_missing_players$player_name)
```

```
## [1] NA NA NA NA
```

These players only appear once and do not have a name so I couldn't match based on ID. I have decided to remove these players from the dataset.

```
#remove players with no name
player_data <- player_data %>%
  filter(!is.na(player_name))
```

We're ready to start the analysis now. Let's delete some of the variables we created that we likely won't use again for cleanliness

```
rm(missing_players)
rm(player_missing_values)
rm(team_missing_values)
rm(updated_missing_players)
```

Question 1

QUESTION: What was the Warriors' Team offensive and defensive eFG% in the 2015-16 regular season? Remember that this is in the data as the 2015 season.

#Let's start with the offensive eFG$

```
#Subset the data to be only warriors data from the 2015 season
warriors_2015 <- team_data %>%
  filter(season == 2015 & (off_team == "GSW" | def_team == "GSW") & gametype == 2)

# Calculate offensive eFG% for Warriors
warriors_offensive_stats <- warriors_2015 %>%
  filter(off_team == "GSW") %>%
  summarise(
    FGM = sum(fgmade),
    FG3M = sum(fg3made),
    FGA = sum(fgattempted)
  ) %>%
  mutate(
    eFG_offensive = ((FGM + FG3M * 0.5) / FGA) * 100
  )
print(warriors_offensive_stats$eFG_offensive)
```

```
## [1] 56.29718
```

```
# Let's do the same for defense
warriors_defensive_stats <- warriors_2015 %>%
  filter(def_team == "GSW") %>%
  summarise(
    FGM = sum(fgmade),
    FG3M = sum(fg3made),
    FGA = sum(fgattempted)
  ) %>%
  mutate(
```

```
    eFG_defensive = ((FGM + FG3M * 0.5) / FGA) * 100
  )
print(warriors_defensive_stats$eFG_defensive)
```

```
## [1] 47.86465
```

ANSWER 1:

Offensive: 56.3% eFG
Defensive: 47.9% eFG

Question 2

QUESTION: What percent of the time does the team with the higher eFG% in a given game win that game? Use games from the 2014-2023 regular seasons. If the two teams have an exactly equal eFG%, remove that game from the calculation.

Since we are interested in the eFG% in ever game, let's add this column to the dataset and compare the eFG%'s to the winner on a gamely basis

```
# created eFG% column using the formula
team_data <- team_data %>%
  mutate(
    eFG = (fgmade + fg3made * 0.5) / fgattempted
  )
# We are interested in only 2014-2023 games
team_data_2014_to_2023 <- team_data %>%
  filter(gametype==2 & (2014 <= season & season<= 2023))

# Group by game so each game is its own row
games_2014_to_2023_efg <- team_data_2014_to_2023 %>%
  group_by(nbagameid) %>%
  summarise(
    home_team = off_team[which(off_home == 1)],
    home_efg = eFG[which(off_home == 1)],
    home_points = points[which(off_home == 1)],
    away_team = off_team[which(off_home == 0)],
    away_efg = eFG[which(off_home == 0)],
    away_points = points[which(off_home == 0)]
  )

# Determine the winner and compare eFG%
games_2014_to_2023_efg <- games_2014_to_2023_efg %>%
  mutate(
    winner = if_else(home_points > away_points, home_team, away_team),
    higher_efg_team = if_else(home_efg > away_efg, home_team, away_team)
  )

# Remove games with equal eFG%
games_2014_to_2023_efg_filtered <- games_2014_to_2023_efg %>%
  filter(home_efg != away_efg)

# Calculate the percentage
```

```r
higher_efg_wins <- games_2014_to_2023_efg_filtered %>%
  summarise(
    total_games = n(),
    higher_efg_wins = sum(winner == higher_efg_team)
  ) %>%
  mutate(
    percent_higher_efg_wins = higher_efg_wins / total_games * 100
  )
print(higher_efg_wins$percent_higher_efg_wins)
```

```
## [1] 81.60141
```

ANSWER 2:

81.6%

Question 3

QUESTION: What percent of the time does the team with more offensive rebounds in a given game win that game? Use games from the 2014-2023 regular seasons. If the two teams have an exactly equal number of offensive rebounds, remove that game from the calculation.

Here, we will take a similar approach to the last question, but instead of looking at eFG%, we will be looking at offensive rebounds

```r
# We already subsetted a df with team data from the 2014-2023 regular seasons
team_data_2014_to_2023_oreb <- team_data_2014_to_2023 %>%
  group_by(nbagameid) %>%
  summarise(
    home_team = off_team[which(off_home == 1)],
    home_oreb = reboffensive[which(off_home == 1)],
    home_points = points[which(off_home ==1)],
    away_team = off_team[which(off_home ==0)],
    away_oreb = reboffensive[which(off_home ==0)],
    away_points = points[which(off_home == 0)]
  )

# Determine which team won and which team had more offensive rebounds
team_data_2014_to_2023_oreb <- team_data_2014_to_2023_oreb %>%
  mutate(
    winner = if_else(home_points > away_points, home_team, away_team),
    higher_oreb_team = if_else(home_oreb > away_oreb, home_team, away_team)
  )

# Remove games with equal orebs
team_data_2014_to_2023_oreb_filtered <- team_data_2014_to_2023_oreb %>%
  filter(home_oreb != away_oreb)

# Calculate the percentage
higher_oreb_wins <- team_data_2014_to_2023_oreb_filtered %>%
  summarise(
    total_games = n(),
    higher_oreb_wins = sum(winner == higher_oreb_team)
```

```
  ) %>%
  mutate(
    percent_higher_oreb_wins = higher_oreb_wins / total_games * 100
  )
print(higher_oreb_wins$percent_higher_oreb_wins)
```

```
## [1] 46.21415
```

46.2% :

46.2%

Question 4

QUESTION: Do you have any theories as to why the answer to question 3 is lower than the answer to question 2? Try to be clear and concise with your answer.

ANSWER 4:

Question 5

QUESTION: Look at players who played at least 25% of their possible games in a season and scored at least 25 points per game played. Of those player-seasons, what percent of games were they available for on average? Use games from the 2014-2023 regular seasons.

For example:

- Ja Morant does not count in the 2023-24 season, as he played just 9 out of 82 games this year, even though he scored 25.1 points per game.
- Chet Holmgren does not count in the 2023-24 season, as he played all 82 games this year but scored 16.5 points per game.
- LeBron James does count in the 2023-24 season, as he played 71 games and scored 25.7 points per game.

ANSWER 5:

XX.X% of games

Question 6

QUESTION: What % of playoff series are won by the team with home court advantage? Give your answer by round. Use playoffs series from the 2014-**2022** seasons. Remember that the 2023 playoffs took place during the 2022 season (i.e. 2022-23 season).

ANSWER 6:

Round 1: XX.X%
Round 2: XX.X%
Conference Finals: XX.X%
Finals: XX.X%

Question 7

QUESTION: Among teams that had at least a +5.0 net rating in the regular season, what percent of them made the second round of the playoffs the **following** year? Among those teams, what percent of their top 5 total minutes played players (regular season) in the +5.0 net rating season played in that 2nd round playoffs series? Use the 2014-2021 regular seasons to determine the +5 teams and the 2015-2022 seasons of playoffs data.

For example, the Thunder had a better than +5 net rating in the 2023 season. If we make the 2nd round of the playoffs **next** season (2024-25), we would qualify for this question. Our top 5 minutes played players this season were Shai Gilgeous-Alexander, Chet Holmgren, Luguentz Dort, Jalen Williams, and Josh Giddey. If three of them play in a hypothetical 2nd round series next season, it would count as 3/5 for this question.

Hint: The definition for net rating is in the data dictionary.

ANSWER 7:

Percent of +5.0 net rating teams making the 2nd round next year: XX.X%
Percent of top 5 minutes played players who played in those 2nd round series: XX.X%

Part 2 – Playoffs Series Modeling

For this part, you will work to fit a model that predicts the winner and the number of games in a playoffs series between any given two teams.

This is an intentionally open ended question, and there are multiple approaches you could take. Here are a few notes and specifications:

1. Your final output must include the probability of each team winning the series. For example: "Team A has a 30% chance to win and team B has a 70% chance." instead of "Team B will win." You must also predict the number of games in the series. This can be probabilistic or a point estimate.
2. You may use any data provided in this project, but please do not bring in any external sources of data.
3. You can only use data available prior to the start of the series. For example, you can't use a team's stats from the 2016-17 season to predict a playoffs series from the 2015-16 season.
4. The best models are explainable and lead to actionable insights around team and roster construction. We're more interested in your thought process and critical thinking than we are in specific modeling techniques. Using smart features is more important than using fancy mathematical machinery.
5. Include, as part of your answer:

- A brief written overview of how your model works, targeted towards a decision maker in the front office without a strong statistical background.
- What you view as the strengths and weaknesses of your model.
- How you'd address the weaknesses if you had more time and/or more data.
- Apply your model to the 2024 NBA playoffs (2023 season) and create a high quality visual (a table, a plot, or a plotly) showing the 16 teams' (that made the first round) chances of advancing to each round.

Part 3 – Finding Insights from Your Model

Find two teams that had a competitive window of 2 or more consecutive seasons making the playoffs and that under performed your model's expectations for them, losing series they were expected to win. Why do you think that happened? Classify one of them as bad luck and one of them as relating to a cause not currently accounted for in your model. If given more time and data, how would you use what you found to improve your model?

ANSWER :